

Defence **MISSILES**



AGNI I

Range: 700 km
Weight: 12 tonnes
Height: 16 metres
Engine diameter: 1 metre
Warhead weight: 1,000 kg
Target: Pakistan-specific missile
Tests Done: Five (all successful)
Status: Inducted into forces



AGNI II

Range: 800-2,000 km
Weight: 17 tonnes
Height: 21 metres
Engine diameter: 1 metre
Warhead weight: 1,000 kg
Target: Pakistan and China
Tests Done: Five (two failures)
Status: Inducted into forces



AGNI III AT THE LAUNCHPAD; THE ARMY CONGRATULATES THE SCIENTISTS (BELOW)

AGNI III

Range: 1,500-3,500 km
Weight: 48 tonnes
Height: 17 metres
Engine diameter: 2 metres
Warhead weight: 1,500 kg
Target: China
Tests Done: Four (one failure)
Status: User Trials

when the missile roared across the firmament and struck its target 3,500 km away. In May 2008, they tested it again and came back with flawless results.

The scientists were nervous for other reasons though. In 2009, two user trials of Agni II, a distant cousin of Agni III with the capability to traverse 2,000 km, failed. This occurred five years after Agni II had been formally inducted into the strategic defence forces in 2004. Agni II was regarded as India's strategic work-horse capable of striking targets in both Pakistan and China. Agni I, another variant with a reach of 700 km, had also been inducted but, given its shorter range, was Pakistan-specific. There was a clear and worrying gap in India's defence preparedness and suddenly much of DRDO's credibility hinged on the success of Agni III.

At precisely 10.50 a.m. on February

7, the Agni III missile fumed like a mythical dragon, letting out clouds of bright orange gas. It then lifted off with a deafening roar, arching gracefully into the sky, leaving a trail of white smoke in its wake. Inside the block-house, the scientists sat glued watching giant television screens that were charting the missile's trajectory on a graph. All systems seemed to be working perfectly as the missile followed a ballistic course, reaching a maximum height of 350 km before descending over the Indian Ocean. It splashed down at a predetermined zone 3,500 km away a little over 13 minutes after take-off. A civilian aircraft would have taken two to three hours to cover the same distance. With sensors and radars confirming its success, a loud cheer broke out in the control room. A smiling Avinash Chander, programme director of Agni, said, "We

have performed a hat-trick. The missile has proved its consistency. The development phase of Agni III is finally over and it is ready for induction."

The test had even greater significance. Apart from improvement in accuracy ensured by multiple sensors, the tests also validated Agni III's ability to intercept satellites if it was ever needed in the future. That was a bonus as China had tested a similar system recently. Summing up the achievement, V.K. Saraswat, scientific adviser to the Defence Minister and DRDO chief, said, "With its successful launch the technology of long range ballistic missiles that we have built has reached a high level of maturity and India now has a credible strategic deterrent against all its adversaries in the region."

Agni III's range of 3,500 km brings both Beijing and Shanghai in range of